Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

Anonymous ACL submission

Abstract

Document parsers for retrieval-augmented generation (RAG) are commonly selected by intrinsic quality metrics—text fidelity, boundary clarity—although a production pipeline ultimately depends on whether the parsed content can be *retrieved*. Building a RAG system over Korean government documents, we find the assumption fails operationally: parser selection alone moves retrieval Hit@1 by 35.2 absolute points ($0.197 \rightarrow 0.549$, a $2.8\times$ relative change), and the parsers with the cleanest chunk boundaries rank near the bottom under retrieval. We introduce the Retrieval-Conditional Parsing Score (RCPS), a training-free selection protocol that ranks the parsers and chunkers a team should deploy using a held-out Q–A probe, averaging over multiple retrievers, with format-invariant answer matching; and a retriever-free coverage diagnostic that attributes missing answers to parser omission versus chunk-boundary splitting before any retriever is run. On our production pipeline 20.2% of gold answers are absent from the parser output—constant across eight chunkers—while at most 2.3% are split, so re-chunking cannot recover the loss; parser choice spans an RCPS range of 0.51 against 0.06 for chunker choice. Controlled cross-domain perturbations (OHR-Bench) confirm the mechanism: semantic noise collapses retrieval while boundary metrics barely move. Finally, we bound parser-side training: retrieval-aware preference optimisation lifts OHR-Bench Hit@5 by +0.85 pp, but a simpler fidelity-distillation control achieves +1.22 pp with overlapping confidence intervals—we find no evidence that a retrieval reward improves over a strong fidelity control. We release KoGovDoc-RAG, a Korean document-RAG benchmark, with a reference implementation of RCPS.

1 Introduction

Retrieval-augmented generation (RAG) is increasingly deployed over real-world document collections—government records, enterprise reports, technical manuals—that exist only as scanned or born-digital PDFs. Every such pipeline begins with a document parser that converts each page to text, which is then chunked, embedded, and retrieved. Because the parser sits at the head of the pipeline its errors cascade downstream; the practical question is how to choose it. Faced with many candidates—OCR engines, layout models, vision–language models—practitioners rank them by intrinsic parsing-quality metrics that reward clean-looking output: low text-edit distance and high boundary clarity (Zhao et al., 2025), on the assumption that cleaner output retrieves better.

We hit this problem building a production RAG system over Korean government documents, and found the assumption fails operationally. A team ranking by intrinsic metrics would deploy MinerU (OpenDataLab, 2025), whose boundary-clarity score (0.72) is matched only by Marker; yet its retrieval Hit@1 is 0.20, against the 0.50–0.55 of three lower-scoring vision–language parsers—35.2 absolute Hit@1 points ($2.8\times$ relative) from parser selection alone. On our parser set, Boundary Clarity ranks the candidates in nearly the opposite order from retrieval (Pearson $r = -0.81$, $n = 5$; sensitive in sign to the corpus), but the robust signal is the mechanism: on a cross-domain English check (OHR-Bench, Zhang et al., 2025a), injecting semantic noise collapses retrieval while Boundary Clarity barely moves, so the intrinsic metric cannot perceive

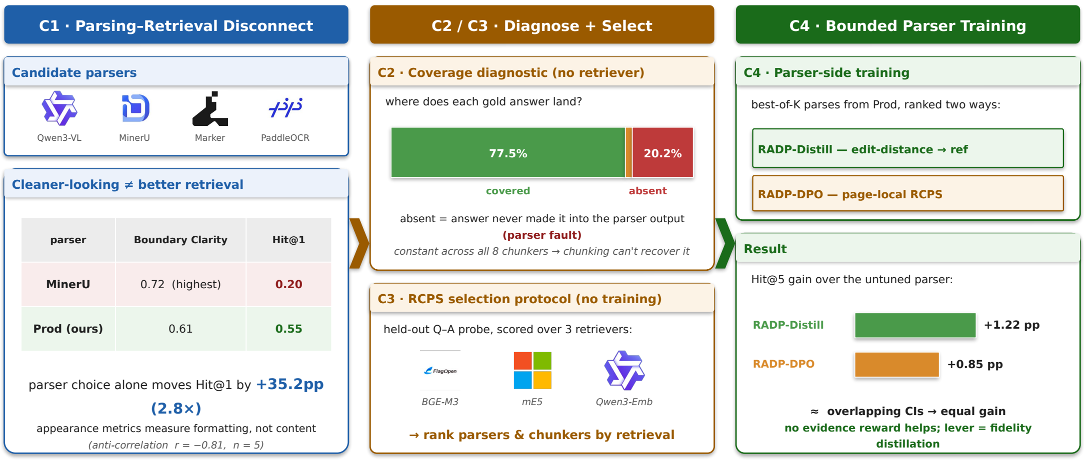

Figure 1: **Overview.** **(C1)** Selecting a document parser by appearance (Boundary Clarity) rather than retrieval is a trap: MinerU has the highest Boundary Clarity yet the lowest Hit@1, and parser choice alone swings Hit@1 by 2.8× (scalar correlations illustrative; §4.2). **(C2)** A retriever-free coverage diagnostic localises the gap to the parser—20.2% of answers are absent from the parser output, constant across all eight chunkers. **(C3)** RCPS selects parsers and chunkers by retrieval over a held-out Q–A probe and three retrievers, with no training. **(C4)** Parser-side training is a bounded lever: fidelity distillation matches the retrieval-reward variant (overlapping CIs)—no evidence the retrieval reward helps.

the content corruption retrieval depends on. Selection by appearance optimises a quantity structurally blind to retrievability—at least for the scanned, layout-heavy documents we study.

The question this paper answers is operational: *given candidate parsers and chunkers, how should a production team pick what to deploy—and is parser training worth the cost?* We answer with RCPS, a training-free protocol that selects the parser and chunker by what retrieval does with their output—this paper's central contribution (C3)—surrounded by the disconnect that motivates it (C1), a retriever-free coverage diagnostic that localises the bottleneck (C2), and an honest bound on what parser-side training adds (C4).

Figure 1 previews all four.

We release **KoGovDoc-RAG**, a Korean document-RAG benchmark of 663 question–answer pairs over 294 pages, together with a reference implementation of RCPS and the RADP-Distill / RADP-aux / RADP-DPO checkpoints.

2 Related Work

Diagnosing the parsing–retrieval gap. OHR-Bench (Zhang et al., 2025a) measures the cascading impact of OCR noise on RAG; EnterpriseDocBench (Singh and Raj, 2026) reports a weak parsing-quality–retrieval correlation; concurrent analyses agree (Sun et al., 2026; Song, 2026). Parsing benchmarks still rank parsers by intrinsic fidelity (Ouyang et al., 2025)—the signal C1 shows is misleading—and vision-based retrieval (Faysse et al., 2025) sidesteps text parsing, orthogonal to the text pipelines we target. None provides a reusable selection protocol or fault localisation—which our RCPS (C3) and coverage diagnostic (C2) supply.

Training-time methods, by pipeline layer. Retrieval-oriented training exists at every layer except the parser: chunking (Günther et al., 2024; Duarte et al., 2024; Zhao et al., 2024, 2025; Singh et al., 2025), contrastive embedding (Conti et al., 2025; LMAR authors, 2025), and retriever- or reader-side tuning (Nguyen et al., 2024; Chia et al., 2025; Yan et al., 2025). Parsers are themselves trained—with RL on edit-distance and layout rewards (Wang et al., 2025) or query-driven for efficient RAG (Wang et al., 2026)—but not on a retrieval signal; we test that signal and find it adds nothing over the fidelity reward such training already uses (C4). Our primary contributions (C1–C3) sit upstream of any training.

3 The RCPS Protocol

3.1 RCPS: Retrieval-Conditional Parsing Score

RCPS ranks parsers by what downstream retrieval does with their output, not by how clean it looks. It is not a new similarity function but an evaluation protocol wrapping ordinary retrieval MRR in three choices, each removing a failure mode our experiments exhibit: (i) it is *extrinsic*, scoring a held-out Q–A probe on the parsed corpus, because intrinsic text metrics reward formatting that retrieval does not depend on (§4.2); (ii) it is *retriever-agnostic*, averaging over several embedders, because a single embedder can silently flip the top-ranked parser (Table 2); and (iii) it uses *format-invariant relevance*, counting a chunk relevant iff its text contains the answer span however formatted—on our grid this shifts scores by 0.02–0.03 without reordering, insurance rather than a ranking driver. The contribution is the protocol—what to measure, on what probe, how to judge relevance—not the scoring function (§4.3).

Given a parser P, a Q–A set $D = \{(q_i, a_i, \text{page}_i)\}$, a set of retrievers R, and cutoffs K, RCPS averages MRR across the cross-product:

$$\text{RCPS}(P) = \frac{1}{|R|\,|K|} \sum_{r \in R} \sum_{k \in K} \text{MRR@}k(r, P, D). \tag{1}$$

We use $R = \{$BGE-M3 (Chen et al., 2024), multilingual-e5-large (Wang et al., 2024), Qwen3-Embedding-8B (Zhang et al., 2025b)$\}$ and $K = \{1, 5, 10\}$. A chunk is relevant iff its source page matches the answer's and the gold span is a substring under whitespace- and markdown-insensitive normalisation. Figure 2 sketches the protocol; the implementation is released.

Running the protocol. A team with m candidate parsers (1) holds out a probe of a few hundred Q–A whose answers occur verbatim in the corpus (ours are LLM-generated and judge-filtered, §4.1); (2) parses the evaluation pages once per candidate; (3) chunks, embeds with the $|R|$ retrievers, and computes Equation 1; (4) deploys the arg-max—m parse passes plus $m\,|R|$ embed-and-retrieve runs, no training, no human relevance labels. Holding the parse fixed instead ranks chunkers with the same probe. RCPS returns a ranking under a fixed probe, not an absolute score.

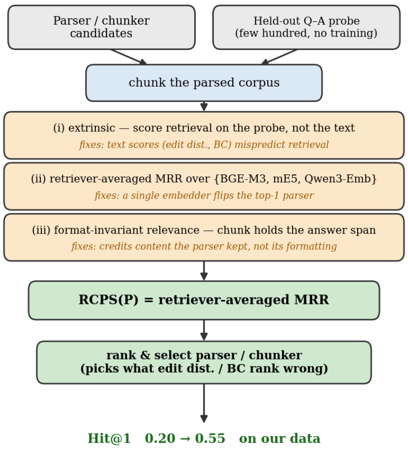

Figure 2: RCPS as a protocol, not a new scoring function (C3): ordinary retrieval MRR wrapped in a held-out Q–A probe, retriever-averaging, and format-invariant relevance. Run with no training, it ranks the parser and chunker a team would deploy.

3.2 Coverage Diagnostic — Locating the Gap (Parser vs Chunker)

RCPS scores the parser, chunker, and retriever jointly, so a low score does not localise the fault. We separate the parser from the chunker with a retriever-free diagnostic: holding the parser output fixed and varying the chunker, we classify each gold answer as *covered* (inside a single chunk, retrievable); *split* (present in the page text but cut across chunks—a chunker fault, recoverable with overlap); or *absent* (not in the parser output at all—a parser fault, unrecoverable by chunking). Coverage is the ceiling on retrieval, since split and absent answers score zero; the split/absent breakdown attributes the gap to a pipeline layer and decides whether a parser-side fix can help (§4.2). Relevance reuses RCPS's format-invariant matching.

4 Experiments

4.1 Setup

We construct **KoGovDoc-RAG**: 663 Q–A pairs over 294 pages of Korean government documents, generated with the OpenAI `gpt-5.4-2026-03-05` model and verified by

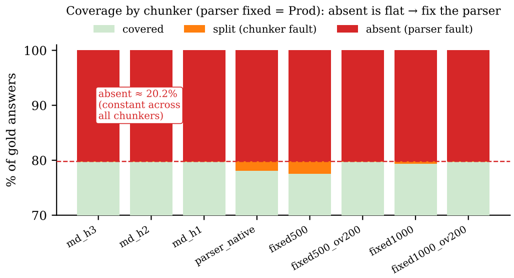

Figure 3: The coverage diagnostic (C2): with the parser fixed (Prod) and the chunker varied, each gold answer is *covered*, *split* (a chunker fault, recoverable with overlap), or *absent* (a parser fault, unrecoverable by chunking). The absent rate is $\approx 20.2\%$ and essentially constant across all eight chunkers, while split is at most 2.3% (§4.2)—chunking cannot close a gap the parser already lost.

an LLM-as-judge against the teacher-distilled reference markdown (Qwen3-VL-30B pseudo-labels; 100 eval Q–A sampled, 94/100 accept). For RADP-aux's full-scale training we additionally generate 6,164 Q–A on the 2,667-page Prod training set. Cross-domain replication uses OHR-Bench (Zhang et al., 2025a) (seven domains, 2,264 verbatim-answerable Q–A; its Law + Manual subset of 1,043 Q–A is used only for the data-mix-sensitivity comparison in §4.2) across the three released parser outputs (gt, MinerU, Qwen2.5-VL) plus twelve controlled noise perturbations.

Throughout, **Prod** denotes our production parser, a Qwen3-VL-2B model (Qwen Team, Alibaba, 2025) fine-tuned for Korean document parsing; all training deltas are measured against it. RADP-aux uses a 73-page held-out fold (202 Q–A); RADP-DPO, SimPO, and the §4.4 analysis use the combined 242-page fold (train $\cup$ eval, 663 Q–A), with preference pairs built only from the 169-page train fold. The same 663 Q–A occupy 242 of the 294 KoGov pages; the coverage diagnostic (§4.2) runs over Prod's full 294-page output, the extra 52 Q–A-free pages acting as distractors. Parses for the 12-variant comparison (§4.4) use deterministic decoding (temperature 0, max 1536 tokens). We fine-tune with LoRA (Hu et al., 2022) ($r = 8$, $\alpha = 32$), sweeping $\lambda \in \{0, 0.1, 0.3, 0.5\}$ for RADP-aux ($\lambda = 0$ drops the contrastive term; the aux recipe itself shifts output from Prod, Table 6, so the sweep is read within the aux family). RCPS uses the three retrievers and cutoffs of §3.1; we report paired percentile-bootstrap 95% CIs (1,000 resamples), preserving Q–A pairing across systems.

4.2 The Parsing–Retrieval Disconnect and Coverage Diagnostic (C1–C2)

Korean government documents (Figure 4; full grid in Table 1, §4.3). RCPS spans 0.07–0.58 across the six parsers—vision–language parsers on top, OCR systems trailing. The 30B teacher ranks first (RCPS 0.584) with the deployable 2B Prod 0.001 behind (0.583), so Figure 4b contrasts Prod against the appearance-ranked MinerU. The intrinsic Boundary Clarity metric (Zhao et al., 2025) anti-correlates with RCPS at Pearson $r = -0.81$ ($n = 5$, excluding PaddleOCR, whose BC is undefined): the two cleanest-boundary parsers (MinerU, Marker; BC ≈ 0.72) land near the bottom, while the lower-BC vision–language parsers retrieve best (dropping the 38-page Marker subset leaves this unchanged, $n = 4$, $r = -0.74$). MoC's Chunk Stickiness is similarly disconnected ($r = +0.26$); neither intrinsic axis predicts RCPS.

Cross-domain: the mechanism. Across all seven OHR-Bench domains (2,264 Q–A) we evaluate 15 parser-output variants—three released outputs plus twelve controlled noise perturbations (Appendix E). Within each semantic-noise family, Boundary Clarity stays flat while RCPS collapses: for the MinerU family BC holds at 0.71–0.74 from clean to severe while RCPS falls −51%, with GOT and Qwen2.5-VL showing the same direction (Table 8). Semantic noise that destroys retrievable content does not lower Boundary Clarity: the intrinsic metric sees formatting, not content.

The aggregate cross-variant correlation is sensitive to the document mix: Pearson BC↔RCPS across all 15 variants is -0.35 on Law + Manual alone but $+0.25$ on the full seven-domain corpus. We therefore report the per-family mechanism, which reproduces in every domain, as the robust finding, and treat the cross-variant scalar as illustrative only.

Locating the gap: parser or chunker? (Figure 3; per-chunker table in Appendix D). The disconnect shows that intrinsic metrics mislead, but not which pipeline layer is at fault. Applying the §3.2 classifica-

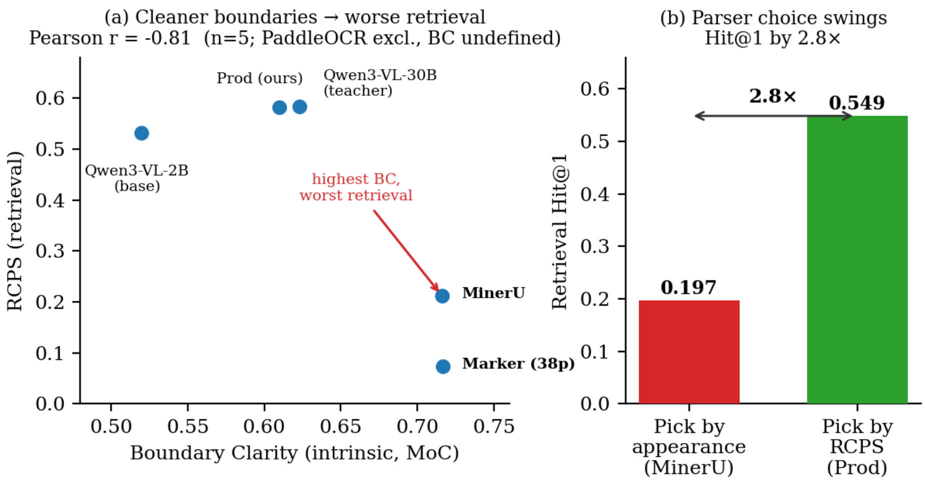

Figure 4: The parsing–retrieval disconnect on Korean government documents. (a) Boundary Clarity (intrinsic, MoC) anti-correlates with RCPS across parsers (Pearson $r = -0.81$, $n = 5$; PaddleOCR excluded as its BC is undefined): the cleanest-boundary parsers MinerU and Marker land near the bottom, while the lower-BC vision–language parsers retrieve best. (b) Choosing the parser by RCPS (Prod) rather than by appearance (MinerU) improves retrieval Hit@1 by 35.2 absolute points (0.197 → 0.549; 2.8× relative).

Parser	BC	CS	RCPS	Hit@1
Qwen3-VL-30B (teacher)	0.623	3.38	**0.584**	0.545
Prod (ours, 2B)	0.610	3.07	0.583	0.549
Qwen3-VL-2B (base)	0.520	3.74	0.532	0.500
MinerU	0.716	2.81	0.212	0.197
PaddleOCR	—	3.46	0.140	0.125
Marker (38p)	**0.717**	3.41	0.073	0.068

Table 1: The KoGov six-parser grid (RCPS averaged over three retrievers; Marker on its 38-page subset). Simultaneously the C1 evidence—BC anti-correlates with RCPS, $r = -0.81$ ($n = 5$; PaddleOCR excluded, BC undefined); CS similarly uninformative ($r = +0.26$; scale differs from Table 6)—and the C3 deliverable: the deployment ranking a team reads off.

tion with no retriever to Prod's output (294 pages, 663 Q–A), 20.2% of answers are absent and at most 2.3% split, with the absent rate constant across all eight chunkers—the boundary-independence a parser fault must show. Roughly one answer in five is never produced, so no re-chunking recovers it: the gap is a parser problem, motivating the training in §4.4.

4.3 RCPS Selects Both Parsers and Chunkers (C3)

Selecting parsers. A selection protocol must separate the alternatives a practitioner actually compares. The six-parser grid of §4.2 (Table 1, Figure 4) is itself an RCPS ranking: it tells a team that Prod (RCPS 0.583, Hit@1 0.55) beats MinerU (0.21, 0.20) by 2.8×, the ordering Boundary Clarity inverts. Both appearance-ranked leaders (MinerU, Marker) sit in the grid's bottom half, and the spread RCPS resolves—0.073 to 0.584—is the full range of deployment outcomes.

Selecting chunkers. On a fixed Prod output, the same probe separates four chunking strategies cleanly—markdown-header (md-h3, RCPS 0.593) > parser-native (0.583) > LumberChunker (Duarte et al., 2024) (0.557) > fixed-size (0.535)—whereas intrinsic boundary metrics rank them inconsistently, or rank fixed-size highest because its boundaries are clean by construction. The chunker spread (0.06) is nearly an order of magnitude smaller than the parser spread (0.51), as the coverage diagnostic predicts: at most 2.3% of answers are chunker-recoverable (§4.2), so chunking moves retrieval only within the band the parser sets. One probe set thus ranks both knobs—and says which is worth turning first.

RCPS is not single-embedder MRR (Table 2). RCPS's value is the operational package—extrinsic, retriever-averaged, format-invariant, no training (§3.1)—not a wholesale reordering: stripped to a single em-

Protocol configuration	Top pair	τ vs RCPS
naive MRR (1 embedder)	Prod $\succ$ 30B	0.87
+ retriever-averaging	30B $\succ$ Prod	1.00
+ format-invariant	Prod $\succ$ 30B	0.87
full RCPS	30B $\succ$ Prod	—

Table 2: RCPS protocol ablation on the KoGov six-parser grid: only the inverted top pair is shown—the lower ranks Qwen3-VL-2B $\succ$ MinerU $\succ$ PaddleOCR $\succ$ Marker are identical throughout. Retriever-averaging alone flips the top pair; format-invariance shifts scores but not order. Kendall τ is over the full ranking; "30B" = the Qwen3-VL-30B teacher.

bedder (BGE-M3), the ranking still agrees at Kendall $\tau = 0.87$. What the single embedder cannot do is resolve the deployment decision at the top: it inverts the near-tied top pair (naive MRR ranks Prod first, full RCPS the 30B teacher; 0.583 vs 0.584), a flip a team trusting its production embedder inherits silently. Format-invariant matching shifts every score by 0.02–0.03 without reordering. We claim the ablation as evidence the protocol choices are not redundant, not as a large ranking change.

4.4 Parser-Side Training Is a Bounded, Reward-Agnostic Lever (C4)

Does training the parser on a retrieval signal help? We test three Retrieval-Aware Document Parsing (RADP) variants—a hidden-state auxiliary loss (RADP-aux), discrete-output DPO on a page-local RCPS reward (RADP-DPO) (Rafailov et al., 2023), and a reward-agnostic best-of-K control ranking candidates by edit distance to the reference markdown (RADP-Distill). Among the retrieval-aware variants only RADP-DPO yields a reliable positive—the auxiliary loss is sub-threshold and reference-free SimPO (Meng et al., 2024) negative—lifting OHR-Bench Hit@5 by $+0.85$ pp $[+0.35, +1.43]$, significant across all seven domains (KoGov $+1.96$ to $+2.11$ pp, exploratory). But the RADP-Distill control matches or exceeds it ($+1.22$ pp $[+0.35, +2.15]$; Table 3), statistically indistinguishable: **we find no evidence that the retrieval reward improves over fidelity-based candidate selection** at the powered endpoint. The lever is text fidelity, not chunking. Recipe, per-variant results, full metric

Variant	Selection signal	ΔHit@5 [95% CI]
RADP-Distill	ref. edit dist.	**+1.22** $[+0.35, +2.15]$
RADP-DPO-R2	page-local RCPS	+0.85 $[+0.35, +1.43]$
RADP-DPO-R3	+ hard negatives	+1.03 $[+0.24, +1.84]$

Table 3: C4 at the powered endpoint: OHR-Bench ΔHit@5 vs Prod, in pp (2,264 Q–A, seven English domains, three-retriever macro, paired bootstrap). The reward-agnostic RADP-Distill control matches or exceeds the retrieval-reward variants, with overlapping CIs. RADP-aux is sub-threshold and SimPO negative on KoGov (Appendix B).

grid (Table 4), and mechanism statistics: Appendices A–C.

5 Discussion and Conclusion

Selection, not training, is the headline. The largest, most certain effect here is not a training method: choosing a parser by retrieval (RCPS) rather than by intrinsic metrics changes Hit@1 by 35.2 absolute points (2.8×, §4.2). Parser-side training is a secondary, bounded lever of about 1 pp Hit@5, and we find no evidence the retrieval reward improves over plain fidelity distillation (§4.4).

Why the diagnosis and the mechanism agree. C1 finds the cleanest-boundary parser retrieves among the worst; §4.4 finds training helps by raising text fidelity. Both follow from splitting "clean" into two axes—Boundary Clarity measures formatting, TextNED measures content fidelity. MinerU is formatting-clean but loses content; training improves fidelity while leaving the formatting signature unchanged (TextNED 0.240 → 0.158, BC 0.630 → 0.641; Table 6)—the same blindness, seen from both ends.

Playbook and conclusion. For practitioners: (1) select parsers with RCPS, not intrinsic metrics—a few hundred held-out Q–A, no training; (2) run the coverage diagnostic—chunker fix if answers are *split*, parser fix if *absent*; (3) if you train, distil toward clean reference text and require any retrieval-aware method to beat that control; (4) gains concentrate on factoid queries—for layout-heavy mixes, tune the chunker or embedder.

Limitations

Single primary language and statistical power. The C1 diagnostic is strongest in Korean ($n = 5$, $r = -0.81$). On English OHR-Bench the aggregate cross-variant scalar is sign-sensitive to document mix ($r = -0.35$ on Law + Manual, +0.25 on the full seven-domain corpus), so directional consistency holds only for the per-family noise mechanism (§4.2), which replicates in every domain—not for the scalar correlation. The English evidence is also built on three real parser outputs plus twelve controlled perturbations rather than fifteen independent parsers. With these sample sizes the correlations are illustrative rather than inferential.

Q–A generation and construct validity. The KoGov reference markdown is itself pseudo-ground-truth distilled from a Qwen3-VL-30B teacher (with manual de-noising of contaminated samples), and the Q–A were LLM-generated and checked by an LLM-as-judge against that reference (94/100 accept on a 100-Q–A stratified sample); neither the reference parses nor the queries were fully human-verified. Synthetic queries and pseudo-reference bound what RCPS can claim absolutely, but the exposure is narrow: our comparative findings—the 2.8× parser-choice swing and every parser/chunker ranking—hold the Q–A set fixed across systems and so are internally valid regardless of how queries were generated, and the confirmatory cross-domain test (§4.4) uses OHR-Bench's own externally curated Q–A. We release the frozen KoGov eval set for audit.

KoGov is exploratory; OHR-Bench is confirmatory. The headline KoGov cell (R2 +1.96 pp, Table 5) has a paired 95% interval of $[-1.06, +5.03]$—two-sided non-significant and selected from a multi-cell scan—so we claim no two-sided significance on KoGov. The confirmatory evidence is the pre-specified OHR-Bench replication (R2 +0.85 pp $[+0.35, +1.43]$, $n = 2{,}264$). A larger KoGov fold ($\geq$ 1,500 Q–A) would be needed for a powered two-sided result.

Candidate pool and untested paradigms. Preference pairs are sampled from Prod itself; an alternative pool (e.g. chosen/rejected pairs from other parsers) might shift the mechanism away from reference-fidelity, which we did not test. We also test only the hidden-state auxiliary loss, discrete-output DPO with a reference policy, and reference-free SimPO; token-level RL, multi-task interleaving, curriculum schedules, and chunk-granularity oracle distillation are untested. Finally, RADP-DPO is a parser-layer intervention; chunker-, embedder-, and reranker-side training on RCPS-graded chunks are complementary directions left to future work.

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A RADP-DPO Milestone Construction

This appendix gives the full recipe for the RADP-DPO preference data of §4.4. Throughout, Prod is the production parser (Qwen3-VL-2B fine-tuned for Korean document parsing), pages come from the 169-page KoGov train fold, and the page-local RCPS reward scores a candidate parse by Equation 1 restricted to that page's questions.

For each train page we sample K alternative parses from the production parser, chunk and score each by a page-local RCPS (the page's questions against the parse's own chunks plus distractors), and form preference pairs (chosen = higher-RCPS parse, gap $\geq$ 5 pp). The reward is sharpened across three milestones: **R1** ($K = 2$ parses, uniform distractors, $\beta = 0.1$) $\rightarrow$ **R2** (a warm-started iterative round, $\beta = 0.05$) $\rightarrow$ **R3** ($K = 14$ parses with a full-corpus hard-negative pool—the other-page chunks closest to the page's queries—widening the candidate-score gap from about 0.05 to 0.56). We train with a LoRA-toggle reference (π_θ = LoRA on, π_{ref} = LoRA off):

$$\mathcal{L}_{\text{DPO}} = -\log \sigma\big(\beta\,[\Delta_\theta - \Delta_{\text{ref}}]\big), \qquad (2)$$

where $\Delta_\bullet = \log \pi_\bullet(c) - \log \pi_\bullet(r)$ for the chosen (c) and rejected (r) parse. The reference-free SimPO control ($\beta = 2.0$, $\gamma = 1.0$) removes the reference policy entirely.

B KoGov DPO Progression, RADP-aux Sweep, SimPO, and Robustness

This appendix reports the training results behind the C4 claim of §4.4. All systems are LoRA fine-tunes of the same production parser Prod (Qwen3-VL-2B); KoGov results use the 242-page / 663-Q–A fold against the same Prod baseline, with Hit@k macro-averaged over the three retrievers BGE-M3, multilingual-e5-large, and Qwen3-Embedding-8B; Table 4 and the robustness paragraph concern OHR-Bench (2,264 English Q–A, seven domains).

On the KoGov fold (10,000-resample paired bootstrap), the three RADP-DPO milestones improve Hit@5 on parser-native chunking over Prod along the reward-sharpening axis (+1.96 to +2.11 pp, all $P[\Delta>0] \approx 0.90$; per-milestone point estimates and CIs in Table 5). At $n = 663$ every two-sided interval spans zero, so these are exploratory: the Hit@5 / parser-native cell was selected from a multi-cell scan over (chunker $\times$ retriever $\times$ k) with no family-wise correction. The reward-agnostic RADP-Distill control, evaluated on the same fold, reaches +2.61 pp and is the headline; its powered confirmation is the pre-specified OHR-Bench replication (+1.22 pp Hit@5, Table 4).

RADP-aux is sub-threshold. The auxiliary loss peaks at $\lambda = 0.1$ (+1–2 pp RCPS) then declines monotonically as the contrastive objective competes with $\mathcal{L}_{\text{parse}}$ over the same LoRA parameters; every Δ-versus-control interval includes zero and the one-sided $P[\Delta>0]$ never approaches 0.90. The hidden-state route does not reach the deployed markdown—only the discrete output does.

SimPO is negative. The reference-free length-normalised variant gives uniformly negative deltas (-0.7 to -1.7 pp Hit@5), indicating the DPO reference policy is load-bearing: without it the parser drifts from the production distribution before any preference signal can compound.

Robustness. The headline RADP-Distill OHR-Bench gain survives two checks. Because it ranks candidates by edit-distance and uses no retriever, it cannot overfit any embedder; its Hit@5 gain is near-uniform across the three (BGE-M3 +1.06, ml-e5-large +1.10, Qwen3-Emb +1.50 pp). And it concentrates on text-evidence (factoid) queries (+1.19 pp) while near-neutral on table-evidence ones (+0.32 pp), consistent with the text-fidelity mechanism.

C Chunk-Level Mechanism Statistics

This appendix asks *why* the C4 training gains of §4.4 arise: whether training moves chunk structure (boundaries, cohesion) or text content. We measure four chunk-level statistics on the 242-page KoGov fold for every trained system and Prod, the shared baseline, in Table 6: MoC Boundary Clarity (BC, adjacent-chunk discontinuity), MoC Chunk Stickiness (CS, within-chunk cohesion), normalised edit distance to the reference markdown (TextNED, character-level Levenshtein distance ÷ longer-string length), and chunking shape.

The signal. All three RADP-DPO milestones reduce TextNED from Prod's 0.240 to 0.163–0.185, a 23–32% move toward the reference markdown; R1 and R2 achieve the largest reductions (0.167, 0.163). RADP-aux instead increases TextNED (0.318–0.626) because its hidden-state objective degrades surface text. TextNED is positively associated with Hit@5 (R3 the one DPO-cluster exception), and RADP-Distill reduces it furthest (0.158) while leading on retrieval—the cleanest demonstration that text fidelity, not the retrieval reward, is the lever.

Cross-domain replication. On all 4,040 English OHR-Bench pages, zero-shot Prod reaches TextNED 0.192—below its 0.240 on Korean, so English leaves less fidelity headroom. R2 reduces it to 0.1894 (a -0.0026 delta, -1.36%, 95% CI $[-0.0043, -0.0010]$, two-sided significant). R3 pushes the English value marginally lower (0.184) yet does not improve KoGov fidelity, which is why R2 is the headline variant.

The non-signal. The MoC chunking signature is unchanged: BC for every DPO variant lands in 0.60–0.66 (Prod 0.63) and CS near 0.47–0.49 (Prod 0.474); chunks-per-page ($\sim$5) and chunk length (300–321) stay close to Prod's. The gain comes from what is parsed, not how chunks are split.

Secondary patterns. The gain concentrates on text-evidence (factoid) queries (+1.19 pp)—where the verbatim answer span

Δ vs Prod (pp)	Hit@1	Hit@5	Hit@10	MRR@10	nDCG@5
RADP-Distill	**+0.88**	**+1.22**	**+1.32**	**+1.01**	**+1.05**
RADP-DPO-R2	+0.53	+0.85	+0.81	+0.70	+0.74
RADP-DPO-R3	+1.31	+1.03	+0.81	+1.17	+1.15

Table 4: OHR-Bench cross-domain Δ vs Prod, in pp (2,264 Q–A over seven English domains; three-retriever macro; 1,000-resample paired bootstrap). The Hit@5 deltas are two-sided significant (95% CIs: RADP-Distill $[+0.35, +2.15]$, R2 $[+0.35, +1.43]$, R3 $[+0.24, +1.84]$). The other columns are monotone re-cuts of the same per-system ranked list, reported only because practitioners use different cutoffs. RADP-Distill leads at the primary metric Hit@5; R3 reaches higher Hit@1 and MRR@10 point estimates but with overlapping CIs—no evidence the retrieval reward helps at the powered endpoint (§4.4).

Variant (vs Prod = 0.6863)	Hit@5	ΔHit@5 [95% CI]	$P[\Delta{>}0]$	ΔRCPS
RADP-DPO-R3	0.7074	+2.11 $[-0.96, +5.13]$	0.913	+1.72
RADP-DPO-R1	0.7069	+2.06 $[-0.96, +5.13]$	0.907	+0.57
RADP-DPO-R2	0.7059	+1.96 $[-1.06, +5.03]$	0.897	+0.47
RADP-SimPO (ctrl)	0.6793	−0.70 $[-3.77, +2.31]$	0.321	−1.56

Table 5: RADP-DPO milestone progression and the SimPO control on parser-native chunking, 242-page / 663-Q–A fold, 10k paired percentile bootstrap, all against the same Prod baseline (Hit@5 = 0.6863). Macro Hit@k averages the three retrievers of §3.1. All three milestones are strong directional positives ($P[\Delta{>}0] \geq 0.90$) but two-sided non-significant at this fold size; RADP-SimPO is uniformly negative. The RADP-Distill headline (KoGov ΔHit@5 +2.61 pp) is in §4.4 and Table 4. Across three Prod seeds the standard deviation of the mean ΔHit@5 is 0.90 pp.

must land inside a chunk—and is near-neutral on table-evidence ones (+0.32 pp). Because RADP-Distill ranks by edit-distance and uses no retriever, its gain is near-uniform across the three embedders, confirming a parser-level effect rather than an embedder artefact. RADP-aux stays sub-threshold because its hidden-state signal reaches the deployed markdown only through diffuse $\mathcal{L}_{\text{parse}}$ back-flow, never localising to the surface tokens the retriever rewards.

D Coverage Diagnostic per Chunker

This appendix tabulates the retriever-free coverage diagnostic of §3.2 applied to the production parser Prod (Qwen3-VL-2B) on KoGovDoc-RAG (294 pages, 663 Q–A): each gold answer is *covered* (inside one chunk), *split* (in the page text but cut across chunks), or *absent* (not in the parser output), classified by text matching only (Table 7).

E OHR-Bench Noise-Family Grid

This appendix tabulates the controlled-noise grid behind the cross-domain mechanism finding of §4.2: twelve OHR-Bench parser-output perturbations (three parser families × four semantic-noise severities), each scored for Boundary Clarity and RCPS over all seven domains (2,264 Q–A, three-retriever average).

Variant	len	c/pg	clen	BC↑	CS↓	TextNED↓
Prod (ref)	1385	4.82	274	0.630	0.474	0.240
RADP-Distill	1597	5.29	291	0.641	—	**0.158**
aux λ=0.0	799	2.96	245	0.553	0.473	0.626
aux λ=0.1	1380	4.05	321	0.652	0.484	0.423
aux λ=0.3	1930	6.73	272	0.470	0.475	0.332
aux λ=0.5	2035	5.82	327	0.556	0.470	0.318
DPO-R1	1606	4.97	311	0.646	0.474	**0.167**
DPO-R2	1610	4.83	321	0.647	0.476	**0.163**
DPO-R3	1514	4.88	300	0.656	0.485	0.185
SimPO	1703	5.31	305	0.601	0.470	0.186

Table 6: Chunk-level mechanism statistics on the 242-page fold (len = parse length, c/pg = chunks per page, clen = chunk length). RADP-Distill attains the lowest TextNED of all (0.158 < R2's 0.163 < Prod's 0.240) with BC unchanged (0.641, within Prod's range), confirming that text fidelity, not chunk structure, is the operative axis (CS was not measured for RADP-Distill, which uses a fixed chunking scheme). Among retrieval-reward variants, R2 has the lowest TextNED (0.163); R3's KoGov TextNED (0.185) does not beat it.

Chunker	covered	split	absent
md_h3	79.8%	0.0%	20.2%
parser_native	78.1%	1.7%	20.2%
fixed500	77.5%	2.3%	20.2%
fixed500_ov200	79.8%	0.0%	20.2%

Table 7: Answer-coverage diagnostic on Prod's output (294 pages, 663 KoGov Q–A; text matching only, no retriever). The absent rate (a parser fault) is constant at 20.2% across all eight chunkers (four representative rows shown), the required boundary-independence check; the split rate (a chunker fault) is at most 2.3% and vanishes under overlap. Chunkers: md_h{1,2,3} = markdown-header splitting at heading depth k; parser_native = the parser's own segmentation; fixed N[_ov$M$] = fixed N-character windows, optionally with M-character overlap.

Family	BC (clean→sev.)	RCPS (clean→sev.)
MinerU	0.735/0.708/0.715/0.712	0.496/0.413/0.351/0.243
Qwen2.5-VL	0.619/0.614/0.616/0.610	0.467/0.462/0.460/0.429
GOT	–/0.634/0.495/0.650	–/0.383/0.338/0.257

Table 8: Per-family Boundary Clarity and RCPS across semantic-noise severity (clean/mild/moderate/severe) on the seven OHR-Bench domains—the source for the §4.2 noise numbers. Within each family BC stays roughly flat while RCPS falls (MinerU −51%). GOT has no clean baseline; the GT output and three formatting-noise variants enter only the aggregate scalar (§4.2) and are in the released grid.